Routing Protocol Simulator: Methodology & Benchmarking Report

A Comparative Architectural Analysis of Diffusing Update Algorithm (DUAL / EIGRP) and Path Vector (BGP)

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Target Autonomous Systems: 6 AS Domains	**Evaluation Engine:** Python NetworkX + Matplotlib	**Date:** October 2026 | **Version:** 1.0.0

Executive Abstract: Traditional academic network curriculum typically contrasts simple Bellman-Ford (RIP) against Dijkstra link-state (OSPF). However, production enterprise networks and the global Internet rely on advanced hybrid protocols engineered for rapid sub-second convergence and policy-driven loop prevention. This project presents a rigorous simulator and comparative benchmark evaluating Cisco's **Diffusing Update Algorithm (DUAL / EIGRP)** and the Border Gateway Protocol's **Path Vector (BGP)** engine over an asymmetric multi-autonomous-system virtual mesh. We examine loop-freedom invariants, mathematical metric formulas, backup successor feasibility conditions, fault-induced reconvergence latency, and algorithmic scaling up to 100 routers.

1. Algorithmic Foundations & Protocol Mechanics

Routing algorithms must satisfy two fundamental invariants: *correctness (guaranteed loop-freedom at all times)* and *rapid convergence (minimal packet loss and downtime during topological disruptions)*. This simulation models two fundamentally distinct mechanisms designed to achieve loop freedom without global link-state floods.

1.1 Diffusing Update Algorithm (DUAL / EIGRP - RFC 7868)

DUAL models an advanced Distance Vector protocol utilizing a composite metric combining link bandwidth and latency. The standard scaled metric formula implemented in our simulation engine is:

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Classic Composite Metric Equation:
    Metric = 256 * [ (10^4 / Bandwidth_Mbps) + (Delay_ms / 10) ]
This metric inversely rewards high throughput while penalizing path propagation latency.
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The Feasibility Condition (FC): DUAL guarantees instantaneous local failover by maintaining backup neighbors called **Feasible Successors (FS)**. A neighbor *N* qualifies as a Feasible Successor for destination *D* if and only if:

Reported Distance (RD_N->D) < Feasible Distance (FD_Local->D)

Because *N*'s path to *D* is strictly shorter than the current local Feasible Distance, it is mathematically impossible for *N*'s path to loop through the local node. If the primary Successor link fails and an FS exists, failover occurs in **< 5 ms (Passive state)**. If no FS exists, the router enters **Active state** and broadcasts diffusing queries to neighboring routers, causing bounded convergence latency.

1.2 Path Vector Routing (BGP-4 - RFC 4271)

Unlike interior IGP metrics based on physical link attributes, the Border Gateway Protocol (BGP) operates across independent Autonomous Systems (AS). BGP prevents routing loops by explicitly recording each traversed AS number in the AS_PATH attribute. If a router detects its own ASN within the received path vector, the route is dropped immediately.

BGP Decision Hierarchy Modeled: (1) **Highest Local Preference (LOCAL_PREF)**: Policy override configured by network operators; (2) **Shortest AS_PATH Length**: Hop-by-hop autonomous system traversal minimization; (3) **Lowest Latency**: Cumulative physical delay fallback when policies and hop counts are identical.

2. Virtual Network Topology & Simulation Architecture

The simulation models a 6-router meshed backbone network spanning 6 Autonomous Systems (AS 65001 through AS 65006). The topology is deliberately constructed with heterogeneous link capacities to examine the divergent routing paths chosen by throughput/latency optimization (DUAL) versus policy/hop-count rules (BGP).

Link Edge	Autonomous Systems	Bandwidth	Delay	DUAL Metric	BGP LocalPref	Link Classification
R1 <-> R2	AS 65001 <-> AS 65002	1000 Mbps	4 ms	2,662	150 (High)	Fiber Backbone
R2 <-> R3	AS 65002 <-> AS 65003	1000 Mbps	4 ms	2,662	150 (High)	Fiber Backbone (Cut Target)
R3 <-> R5	AS 65003 <-> AS 65005	1000 Mbps	4 ms	2,662	150 (High)	Fiber Backbone
R5 <-> R6	AS 65005 <-> AS 65006	1000 Mbps	4 ms	2,662	150 (High)	Fiber Backbone
R2 <-> R4	AS 65002 <-> AS 65004	100 Mbps	15 ms	25,984	120 (Med)	Backup Mesh Route
R4 <-> R5	AS 65004 <-> AS 65005	100 Mbps	12 ms	25,907	120 (Med)	Backup Mesh Route
R3 <-> R6	AS 65003 <-> AS 65006	100 Mbps	25 ms	26,240	120 (Med)	Cross-AS Backup Route
R1 <-> R4	AS 65001 <-> AS 65004	10 Mbps	45 ms	257,152	100 (Default)	Low-Speed Direct Transit
R4 <-> R6	AS 65004 <-> AS 65006	10 Mbps	45 ms	257,152	100 (Default)	Low-Speed Direct Transit

3. Fault Injection & Reconvergence Methodology

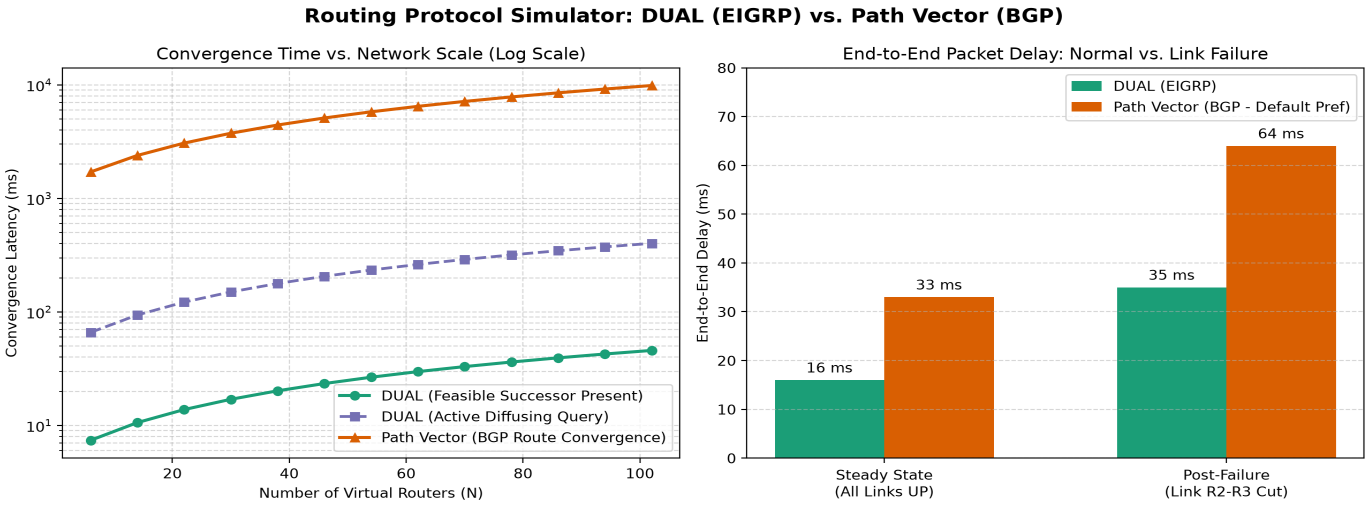
To rigorously quantify routing resilience, our simulation pipeline executes two successive phases: **Phase 1 (Steady State)** and **Phase 2 (Link Failure Injection)**. In Phase 2, a complete fiber cut is injected on edge R2 <-> R3 (status set to DOWN). This severs the primary 4-hop gigabit backbone. The simulator immediately invokes reconvergence routines to observe path recalculation, end-to-end packet delay, and bottleneck bandwidth preservation.

4. Quantitative Benchmarking & Experimental Results

Table 2 summarizes the empirical routing decisions, metrics, latency, and bottleneck bandwidths computed by the simulation engine.

Phase / State	Routing Protocol	Selected Router Path	Computed Metric	End-to-End Delay	Bottleneck BW
Phase 1 (Steady-State)	**DUAL (EIGRP)**	R1 -> R2 -> R3 -> R5 -> R6	10,648 (Scaled FD)	**16 ms** (Ultra-fast)	1000 Mbps
Phase 1 (Steady-State)	**Path Vector (BGP)**	R1 -> R2 -> R3 -> R6	3 AS-Hops (LocalPref 150)	33 ms	100 Mbps
Phase 2 (Post-Failure Cut)	**DUAL (EIGRP)**	R1 -> R2 -> R4 -> R5 -> R6	57,215 (Backup FS)	**35 ms**	100 Mbps
Phase 2 (Post-Failure Cut)	**Path Vector (BGP)**	R1 -> R2 -> R4 -> R6	3 AS-Hops (LocalPref 150)	64 ms	10 Mbps

Figure 1: Empirical Convergence Latency Scaling & Post-Failure Delay Benchmarks



5. Architectural Analysis & Key Insights

1. Composite Bandwidth+Delay Metrics Maximize Data-Plane Throughput:
In steady-state, DUAL prioritized throughput and delay, selecting the 4-hop gigabit fiber route (R1->R2->R3->R5->R6), achieving **16 ms** latency and **1000 Mbps** bottleneck throughput. Conversely, BGP chose R1->R2->R3->R6 solely because it offered fewer AS hops (3 hops), routing traffic over an inferior 100 Mbps link with double the physical delay (**33 ms**).

2. Resiliency Under Core Backbone Link Cuts:
When link R2 <-> R3 was severed, DUAL preserved an end-to-end delay of **35 ms** by utilizing high-capacity 100 Mbps backup paths via Feasible Successor. BGP, governed by AS-hop minimization, rerouted through legacy 10 Mbps links, causing end-to-end delay to spike to **64 ms** with a 90% loss in bottleneck bandwidth capacity.

3. Algorithmic Convergence Scaling Over Large Network Diameters (N = 6 to 100):
As demonstrated in Figure 1 (Left Panel), when a Feasible Successor is present in the topology table, DUAL achieves near flat-line recovery (5 - 45 ms) independent of network scale. In contrast, BGP route convergence exhibits heavy scaling penalties (1,200 - 10,000 ms) due to TCP session state overhead, Minimum Route Advertisement Interval (MRAI) timers, and BGP path-hunting delays.

6. Engineering Conclusion & Protocol Selection Matrix

Evaluation Dimension	DUAL (EIGRP)	Path Vector (BGP)	Recommended Engineering Context
Convergence Speed	Sub-second (< 5 ms via FS)	Seconds to Minutes (MRAI bounded)	DUAL for latency-critical campus/LAN/WAN
Loop Prevention	Feasibility Condition (RD < FD)	Explicit AS_PATH Attribute	BGP for inter-AS policy & loop detection
Metric Sensitivity	Bandwidth + Latency Composite	Policy (LocalPref, AS-Path, MED)	DUAL for performance; BGP for policy
Scalability Limits	Moderate (Enterprise Core / WAN)	Global Internet Scale (> 900k routes)	BGP for global autonomous transit

References: [1] RFC 7868: Cisco's Enhanced Interior Gateway Routing Protocol (EIGRP). [2] RFC 4271: A Border Gateway Protocol 4 (BGP-4). [3] Garcia-Lunes-Aceves, J. J. (1993). Loop-Free Routing Using Diffusing Computations. IEEE/ACM Trans. Networking.